

RIFAT ERDEM SAHIN

FORWARD DEPLOYED ENGINEER — EMBEDDED DELIVERY | AI & PLATFORM ENGINEERING

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PROFESSIONAL SUMMARY

Forward Deployed Engineer with **12+ years** embedding directly inside customer environments to turn ambiguous business problems into shipped, production software — fast. Proven track record parachuting into Fortune 500, finance, defense, and healthcare accounts to build the first working prototype in days, then hardening it into a system the customer's own team can run. Full-stack across AI/LLM integration, data pipelines, cloud infrastructure, and DevSecOps, with the client-facing judgment to translate a stakeholder's vague pain point into an architecture, a demo, and a deployment plan in the same week. Comfortable being the engineer, the architect, and the trusted advisor in the room at once — holds active **UK SC** and **NATO** security clearances for regulated and defense-adjacent engagements.

CORE COMPETENCIES

Embedded Delivery & Rapid Prototyping

- On-site/embedded engineering with enterprise and government customers — requirements to working prototype in days, not sprints
- Rapid iteration loops directly with end users; demo-driven development that keeps the customer in the feedback loop
- Translating ambiguous, non-technical asks from operators and executives into concrete technical scope
- Building the “golden path” integration that becomes the reference implementation for the customer's own engineers

AI & LLM Integration

- Production LLM application development: RAG pipelines, vector databases, prompt engineering, agentic workflows
- AI gateway and multi-model orchestration (OpenAI-compatible APIs, semantic routing, cost-aware model selection)
- AI security: prompt-injection prevention, data privacy controls, PII redaction in customer-facing systems

- watsonx, OpenAI API, Anthropic Claude API, LangChain, LlamaIndex

Full-Stack Platform Engineering

- End-to-end ownership: data ingestion → pipeline → API → UI, deployed and demoed without a hand-off
- Infrastructure as Code (Terraform, Ansible) so prototypes ship as reproducible, customer-owned infrastructure
- Container orchestration with Kubernetes and Docker at scale, across on-prem, hybrid, and multi-cloud
- CI/CD pipeline design so the customer's team can extend the solution after the engagement ends

Regulated & Mission-Critical Environments

- Active **UK SC** and **NATO** security clearances — cleared for defense and government-adjacent deployments
- Zero-downtime migrations for mission-critical financial and government workloads
- Compliance-aware delivery: security review, audit trails, and data-governance built into the first prototype, not bolted on later
- Background-checked: Watchdog (2024), Sterling (2019)

Customer-Facing Technical Leadership

- Sole technical point of contact for enterprise stakeholders during embedded engagements
- Runs discovery workshops, technical demos, and architecture reviews directly with client teams
- Builds internal champions by leaving customers with documentation, runbooks, and a team that can self-serve
- Track record shaping enterprise architecture decisions from inside the customer's own org

KEY ACCOMPLISHMENTS

2025 | IBM | London, UK

Embedded Hybrid Cloud & AI Transformation

- Embedded with Fortune 500 client teams to architect a hybrid cloud transformation platform on IBM Cloud + Red Hat OpenShift, working from the customer's own backlog and constraints
- Delivered watsonx AI automation prototypes on-site, cutting the customer's operational overhead by **35%** within the engagement window
- Led DevSecOps transformation embedded with client infrastructure teams, standing up Ansible Automation Platform and Terraform IaC across multi-cloud environments
- Delivered zero-downtime Kubernetes migrations directly alongside customer engineers for mission-critical financial and government workloads

- **Keywords:** Forward Deployed, Embedded Delivery, Hybrid Cloud, AI Automation, DevSecOps, OpenShift, watsonx
- **Technologies:** IBM Cloud, Red Hat OpenShift, watsonx AI, Ansible, Terraform, Kubernetes, GitHub Actions

2024 | Goldman Sachs | Muscat, Oman

On-Site AI-Driven CI/CD Framework

- Deployed on-site with the trading technology team to co-build an AI-driven CI/CD framework, increasing deployment frequency by **300%**
- Worked directly with client engineers to embed the framework into their existing workflows, cutting operational costs by **30%**
- **Keywords:** Forward Deployed, Client-Embedded, AI, CI/CD, Cost Optimization, Finance
- **Technologies:** Python, AI/ML, Jenkins, GitLab, Kubernetes

2023 | Ypsomed | Switzerland

Embedded IoT Systems Migration

- Embedded with the customer's device and platform teams to migrate IoT systems and optimize workflows in a regulated healthcare environment
- Co-designed the new versioning workflow with client engineers, achieving a **40%** reduction in versioning conflicts and leaving the team self-sufficient
- **Keywords:** Forward Deployed, IoT, Healthcare, Regulated Environment, Migration
- **Technologies:** Git, CI/CD, Azure, IoT Hub

2022 | Cushman & Wakefield | London, UK

Embedded Data Pipeline Delivery

- Sat inside the analytics team to automate ETL processes end-to-end, increasing data processing speed by **50%**
- Shipped and handed over real-time data pipelines the customer's own analysts could operate without further support
- **Keywords:** Forward Deployed, ETL, Data Engineering, Real Estate
- **Technologies:** Python, Apache Airflow, SQL, Azure Data Factory

2021 | Emerson | USA

On-Site Kubernetes & GPU Optimization

- Embedded with manufacturing engineering teams to re-platform ML workloads onto Kubernetes and GPU-based containers
- Improved computational efficiency by **45%**, working iteratively on the factory floor with the customer's own operators

- **Keywords:** Forward Deployed, Kubernetes, GPU, Manufacturing, ML
 - **Technologies:** Kubernetes, Docker, NVIDIA GPU, Python
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TECHNICAL SKILLS

Delivery Model: Embedded/on-site engineering, rapid prototyping, discovery workshops, technical demos, customer handoff & enablement **AI & Machine Learning:** LLM application development, RAG architecture, vector databases (Pinecone, Weaviate, ChromaDB), semantic routing, prompt engineering, AI security & governance, LangChain, LlamaIndex, OpenAI API, Anthropic Claude API, watsonx **DevOps & Infrastructure:** CI/CD pipeline design, Infrastructure as Code, container orchestration, GitOps workflows, monitoring & observability **Cloud Platforms:** Azure (AKS, Azure DevOps, Functions, Data Factory), AWS (EKS, Lambda, S3, CloudFormation), GCP (GKE, Cloud Functions, BigQuery), IBM Cloud, Red Hat OpenShift **Tools & Technologies:** Docker, Kubernetes, Helm, Terraform, Ansible, Jenkins, GitLab CI, GitHub Actions, Prometheus, Grafana, ELK Stack, Apache Airflow, Apache Kafka **Languages:** Python, Bash, PowerShell, Go, YAML, SQL **Version Control:** Git, GitHub, GitLab, Bitbucket

CERTIFICATIONS & EDUCATION

🏠 **Claude Architect Professional Certification** (Anthropic) 🏠 **Microsoft Certified Architect in Cloud Solutions** (70-532) 🏠 Azure Certifications (Various) 🏠 AWS Certifications 🏠 Kubernetes Certifications 🏠 DevOps & Agile Certifications

🎓 **Bachelor of Science**, Southern New Hampshire University, USA 🇺🇸 (2013)

SECURITY CLEARANCES

🔒 **UK SC (Security Check)** — Valid until 2028 🏠 **NATO** — Valid until 2029 ✓ Background checks: Watchdog (2024), Sterling (2019)

WHY FORWARD DEPLOYED ENGINEERING

Forward deployed work is the fastest feedback loop in software: sit with the customer, see the real constraint, ship something that works against it that same week. Twelve years of embedding inside banks, defense-adjacent programs, manufacturers, and healthcare device makers has taught me that the hardest part is rarely the code — it's translating a stakeholder's half-formed problem into a system their own team can trust and run after I leave. I want to keep doing that: land, prototype, harden, hand off.

References and detailed engagement case studies available on request.